

Software Engineering

BRANCH: INFORMATION TECH (IT)

COURSE CODE: IT202

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Software Engineering (IT202) under the Information Tech (IT) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Process Models

Software lifecycle models: Waterfall, Spiral, Prototype, Incremental, RAD. Agile development: Scrum framework, User stories, Kanban.

Unit 2: Requirements & Design

SRS document, Feasibility study. UML diagrams: Use Case, Class, Sequence, Activity diagrams. Architectural styles, modularity, coupling, and cohesion.

Unit 3: Coding & Testing

Coding standards, code refactoring. Software testing: Black-box (Equivalence partitioning, Boundary value analysis) vs White-box (basis path, control flow testing). Unit, Integration, System testing.

Unit 4: Project Management & DevOps

COCOMO estimation model. Risk management, software configuration management. Introduction to CI/CD, Git, automated testing.

*COCOMO: $Effort = a * (KLOC)^b$. Coupling should be minimized (loose); Cohesion should be maximized (high).*

CRITICAL FORMULA / PRINCIPLE:

Key Formulas & Exam Pointers



